

Look-Ahead Bias: What it Means, How it Works

 investopedia.com/terms/l/lookaheadbias.asp

What Is the Look-Ahead Bias?

Look-ahead bias occurs by using information or data in a study or simulation that would not have been known or available during the period being analyzed. This can lead to inaccurate results in the study or simulation. More importantly, a look-ahead bias can unintentionally sway simulation results closer into line with the desired outcome of the test. This leads to economists and analysts putting too much confidence in their models and the ability of the model to predict and mitigate future events. Investors also need to be aware of the potential for look-ahead bias when evaluating particular trading strategies using past data.

Key Takeaways

- Look-ahead bias is when data that was not readily available at the time is used in a simulation of that time period.
- A look-ahead skews the results and leads to overconfidence in models and other frameworks built out of the skewed results.
- A backtested simulation with a look-ahead bias will not show an accurate result. Therefore, careful research is necessary to determine what data was available at the time.

Understanding Look-Ahead Bias

Look-ahead bias often happens in "could have" scenarios, where an investor or other professional considers what is a missed opportunity in hindsight. What that person fails to realize is that they know more now looking back than they did at the time they made the decision. Therefore, it may be unwise to judge their—or others—past performance too harshly in retrospect, especially if key information was missing.

If an investor is backtesting the performance of a trading strategy, it is vital that they only use information that would have been available at the time of the trade to avoid a look-ahead bias. For example, if a trade is simulated based on information that was not available at the time of the trade—such as a quarterly earnings number that was released a month later—it will diminish the accuracy of the trading strategy's true performance and potentially bias the results in favor of the desired outcome.

The Look-Ahead Bias and Other Biases in Investing

Look-ahead bias is one of many biases that must be accounted for when running simulations. Other common biases are sample selection bias, time period bias, and survivorship bias. All of these biases have the potential to sway simulation results closer

into line with the desired outcome of the simulation, as the input parameters of the simulation can be selected in such a way as to favor the desired outcome.

As mentioned, these biases are most clearly seen when investors look back upon the year. Stocks that have performed well throughout the year may now be overbought on the assumption that they will do the same thing the following year. While past performance does influence future performance, it is important for investors to look at the fundamentals of the company carefully as there is always the risk of overvaluation.

If you took the top performing stocks at the end of the year and then tried to choose common data points they had at the start of the year, such as the trailing P/E ratio range, you'd be falling prey to a look-ahead bias because you'd only be looking at stocks you know enjoyed significant growth rather than at all stocks with a similar trailing P/E ratio range at that time. By not including the full range of stocks, you would end up with overconfidence in trailing P/E ratio as the key measure to predict future appreciation. This look-ahead bias can be corrected by widening the sample to all stocks that fit your particular criteria at the start of the year and tracking their outcomes as well.